



INSTRUMENT DOCUMENTATION

LIA (MIL): the complete item bank and key

All 305 items of the instrument in Spanish and English, with the correct answer to each and the exact orientation of every figure in the Visual Rotation subtest

Prepared for	TIMS International
Prepared by	NexaDev LLC
Instrument	LIA — recorded internally as MIL
Contents	305 items · 15 practice and 290 scored
Source	lia-question-bank.json, lia-verbal-en.json and LiaAnswerScoring.cs
Method	Mechanical extraction; the key was recomputed with the production logic

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SECTION 01

Summary

What this document is and where it comes from

This document gathers all **305 items** of the LIA instrument together with the correct answer to each, in Spanish and English. It is for administering the test, checking the scoring, working on the translation and auditing. No item was transcribed by hand: every one comes from the instrument's own data files.

All 305 items	Data/lia-question-bank.json
English verbal text	Data/lia-verbal-en.json
Answer key	Recomputed with LiaAnswerScoring.cs
Counts and timings	SUBTEST_CONFIG in liaService.ts
Figure orientation	VisualRotationItem.tsx

How the key was obtained

Four of the five subtests **store no answer at all**: the correct one is computed from the item's own data at the moment of scoring. That is why the key in this document had to be regenerated rather than read. The production logic was reimplemented exactly as it stands in `LiaAnswerScoring.cs` and checked against worked examples from every subtest. Only Verbal Reasoning stores its own key, and that is reproduced verbatim.

Editing an item changes its answer, silently

Because the key is derived from the item's data, changing a letter or a number moves the correct answer. There is no key file that can fall out of date — but equally there is none that would give the mistake away. After any change to the bank this document is **regenerated**; it is never edited by hand.

SECTION 02

The instrument

The five subtests, timings and answer format

#	Subtest	Español	Practice	Scored	Time	Answer
1	Pattern Recognition	Reconocimiento de Patrones	3	60	3 min	0–4
2	Verbal Reasoning	Razonamiento Verbal	3	50	4 min	A / B / C
3	Numerical Speed	Velocidad Numérica	3	60	4 min	A / B / C
4	Working Memory	Memoria de Trabajo	3	60	4 min	left / right
5	Visual Rotation	Rotación Visual	3	60	5 min	0–3
	Total		15	290	20 min	

The subtests are administered in that order. Each is preceded by its instructions and by three practice items that do not count. The clock runs per subtest, not per session, and a subtest cannot be resumed once its time has run out.

SECTION 03

How it is scored

The exact rule for each subtest

Subtest	Rule
Pattern Recognition	Two rows of four letters. The two letters in each column are compared ignoring case . The answer is the number of matching columns , from 0 to 4.
Verbal Reasoning	Two premises and a question with three options. The key is stored with the item and is given as the option letter, A, B or C, in the order the options are presented.
Numerical Speed	Three numbers. The median is taken, and the number lying farthest from it in absolute distance is sought. The answer is that number's original position : A, B or C. Ties resolve to the earliest position , in the order A, B, C.
Working Memory	Three letters. Each is converted to its position in the alphabet (A=1 ... Z=26) and the distance of each outer letter from the middle one is compared. The answer is left or right , whichever lies farther. A tie resolves to the left .
Visual Rotation	Two rows of three figures, each a letter R either normal or mirrored, and rotated. Two figures match if they share the same handedness : both normal, or both mirrored. The answer is the number of matching columns , from 0 to 3.

Rotation is a distractor, not part of the answer

Visual Rotation is scored **on handedness alone**. The angle of rotation is read and discarded: R_90 and R_270 count as matching, because both are normal Rs. The angles exist to make the judgement harder, not to change the answer. This is the most misread rule in the instrument, which is why the on-screen instruction says “rotations allowed, mirrors not”.

SECTION 04

Languages

What is translated and what only looks like it

Only **Verbal Reasoning** has content that depends on language, and it is fully translated: the 3 practice items and the 50 scored ones exist in Spanish and English, human-reviewed. The other four subtests are **language-neutral by construction** — their items are letters, numbers and figures, so the same item serves both languages. With **one caveat**: Working Memory has neutral items but measures over the **alphabet**, and the alphabet does depend on the language. See the caution at the end of this section.

Subtest	Item content	On-screen instruction
Pattern Recognition	Language-neutral	Spanish and English
Verbal Reasoning	Translated — Spanish and English	No separate instruction: the item's own question does that job, and it is translated
Numerical Speed	Language-neutral	Spanish and English
Working Memory	Neutral items, but it measures over the alphabet	Spanish and English
Visual Rotation	Language-neutral	Spanish and English

Fixed on 24 September 2026

Until that date the four language-neutral subtests carried their instruction hard-coded in Spanish, with no translation branch, as did the countdown warnings: a student working in English saw a Spanish instruction on four of the five subtests. The cause was not a decision but incomplete wiring — the components that draw the question never received the language, although everything above them did. **This is now fixed**: the English instructions printed in this document are the ones the product shows. A **native-speaker review** of those wordings is still outstanding before they can be considered final.

The rest of the platform

Measured on 24 September 2026: the platform has **8 namespaces in English and Spanish, with 4,355 keys in each language at exact parity** — no translation is missing in either direction. Of 617 components, only 7 carried hard-coded Spanish, and all 7 were LIA's. Put another way: LIA was the only hole, and it is now closed. One honest caveat remains: parity proves the translation *exists*, not that it is well written.

Adding Portuguese is five jobs, not one

The application ships Spanish and English only, and a student whose browser is set to Portuguese is resolved to **English** with no warning at all. But “adding Portuguese” is not one task. Each subtest costs something different, and **only two of the five** force a re-norm. Stating it in the aggregate, in either direction, leads to the wrong decision.

Subtest	What Portuguese would need	Re-norm?
Pattern Recognition	The instruction line only. Items are Latin letters, unchanged.	No
Numerical Speed	The instruction line only. Items are integers, unchanged.	No
Visual Rotation	The instruction line only. Items are rotated R figures, unchanged.	No
Working Memory	The instruction line and an item review — see the caution below.	Yes, if items change
Verbal Reasoning	53 items authored afresh in Portuguese. The item is the difficulty.	Yes

Working Memory is not as language-neutral as it looks

Its construct measures **distance in the alphabet**, and the alphabet depends on the language. The scorer uses A=1 ... Z=26.

- In **Spanish** (RAE, 27 letters, with Ñ between N and O): **1 of the 63 scored items** resolves differently for a student reciting the Spanish alphabet. On **M P T** the key is **right**; counting Ñ, the two distances come out equal.
- In **Portuguese as taught before the 1990 accord** (23 letters, no K, W or Y): **19 of the 63 items** use a letter that alphabet does not contain, and of the 44 that remain **3 resolve differently**.

The first **already affects the Spanish now in the field**, not just a hypothetical Portuguese. The second says that for Portuguese this subtest needs an item review, not just a translated instruction.

SECTION 05

Pattern Recognition

Reconocimiento de Patrones · 3 practice + 60 scored · 3 minutes

Instruction · Spanish	¿Cuántas columnas tienen letras iguales? (sin importar mayúsculas/minúsculas)
Instruction · English	How many columns contain the same letter? (case is ignored)

The key is the number of columns whose two letters match, ignoring case (0–4). The **Matches** column says which columns match.

Item	Row 1	Row 2	Key	Matches	Item	Row 1	Row 2	Key	Matches
P1	D F H R	a g t r	1	col. 4	30	D N B P	w n b p	3	col. 2, 3, 4
P2	q a l b	Q A L B	4	col. 1, 2, 3, 4	31	D P G W	d j n r	1	col. 1
P3	M N D E	w f t h	0	none	32	j b q n	J B Q N	4	col. 1, 2, 3, 4
1	q p d t	Q P D T	4	col. 1, 2, 3, 4	33	G K T B	w k m n	1	col. 2
2	m n f q	K N F Q	3	col. 2, 3, 4	34	b m g r	B F T R	2	col. 1, 4
3	R W M Q	j w k h	1	col. 2	35	T N F G	t j f q	2	col. 1, 3
4	T B W J	n b h m	1	col. 2	36	T R G D	t p b d	2	col. 1, 4
5	b k t r	P K T N	2	col. 2, 3	37	b t n w	D M Q K	0	none
6	h m t b	F Q J K	0	none	38	g p h f	T M H R	1	col. 3
7	k d f g	K M H T	1	col. 1	39	d r q j	D R Q J	4	col. 1, 2, 3, 4
8	J F R T	j d b t	2	col. 1, 4	40	R G Q W	t j b w	1	col. 4
9	M N B P	m n b r	3	col. 1, 2, 3	41	m f h b	M F H B	4	col. 1, 2, 3, 4
10	M R N D	g r q h	1	col. 2	42	R F N W	r f g t	2	col. 1, 2
11	t k j g	W K N G	2	col. 2, 4	43	P R J Q	g m n k	0	none
12	k f h m	G J R M	1	col. 4	44	W N B K	w n b k	4	col. 1, 2, 3, 4
13	R H W P	b g k j	0	none	45	d r n q	P B N K	1	col. 3
14	b w n j	H W T J	2	col. 2, 4	46	B R G W	f r m w	2	col. 2, 4
15	w r b d	W R J D	3	col. 1, 2, 4	47	h g b p	K R B P	2	col. 3, 4
16	j t k p	J T K P	4	col. 1, 2, 3, 4	48	t p m g	T P M G	4	col. 1, 2, 3, 4
17	R J T H	r j k h	3	col. 1, 2, 4	49	T P F B	h q n b	1	col. 4
18	W P T R	w q f g	1	col. 1	50	Q R K P	q r k p	4	col. 1, 2, 3, 4
19	G R J D	t r j p	2	col. 2, 3	51	r n j k	R N J Q	3	col. 1, 2, 3
20	J B K D	j b k d	4	col. 1, 2, 3, 4	52	Q R F H	j n p d	0	none
21	B P M F	n p q g	1	col. 2	53	D J B F	d q b f	3	col. 1, 3, 4

Item	Row 1	Row 2	Key	Matches	Item	Row 1	Row 2	Key	Matches
22	H Q P W	m k t f	0	none	54	g w q n	G R J P	1	col. 1
23	P J B M	p n b m	3	col. 1, 3, 4	55	d r b h	D R B W	3	col. 1, 2, 3
24	n k p b	H R D G	0	none	56	F B R T	f b r t	4	col. 1, 2, 3, 4
25	W P F H	w p f h	4	col. 1, 2, 3, 4	57	J Q G R	j q g r	4	col. 1, 2, 3, 4
26	g q b f	K Q B F	3	col. 2, 3, 4	58	B Q J M	b f w m	2	col. 1, 4
27	p d b g	T F K N	0	none	59	m j b f	H K T P	0	none
28	g q h j	G N H J	3	col. 1, 3, 4	60	H W G Q	k m f t	0	none
29	w t d k	W T D P	3	col. 1, 2, 3					

SECTION 06

Verbal Reasoning

Razonamiento Verbal · 3 practice + 50 scored · 4 minutes

The key is the option letter — **A**, **B** or **C** — and beside it the **correct answer written out**, so nobody has to count the options.

One item has a different key in each language

In **practice item 2** the order of the English options differs from the Spanish original, so the key is **B in Spanish** and **C in English**. The engine applies that one exception and no other. No scored item differs: all 50 share a key across both languages.

Practice items (not scored)

Item	Español	English	ES	Answer	EN	Answer
P1	Raquel es más alta que Belén. Belén es más alta que Julia. <i>¿Quién es la más alta?</i> A Raquel · B Belén · C Julia	Anna is taller than Mary. Mary is taller than Olivia. <i>Who is the tallest?</i> A Anna · B Mary · C Olivia	A	Raquel	A	Anna
P2	Miguel es más rápido que Juan. Juan es más rápido que Jesús. <i>¿Quién es el más lento?</i> A Miguel · B Jesús · C Juan	Liam is faster than James. James is faster than Henry. <i>Who is the slowest?</i> A Liam · B James · C Henry	B	Jesús	C	Henry
P3	Víctor es más rico que David. Daniel es más pobre que David. <i>¿Quién es el más pobre?</i> A Víctor · B David · C Daniel	Leo is nicer than David. Owen is meaner than David. <i>Who is the meanest?</i> A Leo · B David · C Owen	C	Daniel	C	Owen

Scored items

Item	Español	English	ES	Answer	EN	Answer
1	Juan es más lento que Jesús. Juan es más rápido que Pedro. <i>¿Quién es más lento?</i> A Juan · B Jesús · C Pedro	Mateo is slower than Liam. Mateo is faster than Pedro. <i>Who is the slowest?</i> A Mateo · B Liam · C Pedro	C	Pedro	C	Pedro
2	Laura es más baja que Paula. Laura es más alta que Eva. <i>¿Quién es más alta?</i> A Laura · B Paula · C Eva	Olivia is shorter than Emma. Olivia is taller than Sophia. <i>Who is the tallest?</i> A Olivia · B Emma · C Sophia	B	Paula	B	Emma
3	Silvia es más inteligente que Lourdes. Silvia es más torpe que Rocío. <i>¿Quién es más inteligente?</i> A Silvia · B Lourdes · C Rocío	Lily is smarter than Mia. Lily is more clumsy than Maria. <i>Who is the smartest?</i> A Lily · B Mia · C Maria	C	Rocío	C	Maria

Item	Español	English	ES	Answer	EN	Answer
4	Pablo está más triste que Luis. Pedro está más triste que Pablo. <i>¿Quién está más feliz?</i> A Pablo · B Luis · C Pedro	Pablo is less happy than Luis. Pedro is less happy than Pablo. <i>Who is the happiest?</i> A Pablo · B Luis · C Pedro	B	Luis	B	Luis
5	Aníbal tiene más frío que Carlos. David tiene más frío que Aníbal. <i>¿Quién tiene más frío?</i> A Aníbal · B Carlos · C David	Lucas is colder than Carlos. David is colder than Lucas. <i>Who is the coldest?</i> A Lucas · B Carlos · C David	C	David	C	David
6	Ana es más baja que Sara. Sara es más baja que Lucía. <i>¿Quién es la más baja?</i> A Ana · B Sara · C Lucía	Anna is shorter than Sara. Sara is shorter than Ava. <i>Who is the shortest?</i> A Anna · B Sara · C Ava	A	Ana	A	Anna
7	Jesús es más inteligente que Mario. Mario es más inteligente que Pablo. <i>¿Quién es el más torpe?</i> A Jesús · B Mario · C Pablo	Evan is smarter than Mario. Mario is smarter than Pablo. <i>Who is the clumsiest?</i> A Evan · B Mario · C Pablo	C	Pablo	C	Pablo
8	Pilar es más lenta que Raquel. Raquel es más lenta que Julia. <i>¿Quién es la más rápida?</i> A Pilar · B Raquel · C Julia	Sofia is slower than Anna. Anna is slower than Emma. <i>Who is the fastest?</i> A Sofia · B Anna · C Emma	C	Julia	C	Emma
9	Belén es más afectuosa que Marta. Isabel es más afectuosa que Belén. <i>¿Quién es más afectuosa?</i> A Belén · B Marta · C Isabel	Charlotte is more caring than Martha. Mia is more caring than Charlotte. <i>Who is the most affectionate?</i> A Charlotte · B Martha · C Mia	C	Isabel	C	Mia
10	Aníbal es más alto que Ángel. Sergio es más bajo que Ángel. <i>¿Quién es el más bajo?</i> A Aníbal · B Ángel · C Sergio	Jack is taller than Owen. Mike is shorter than Owen. <i>Who is the shortest?</i> A Jack · B Owen · C Mike	C	Sergio	C	Mike
11	Elena es más alta que Raquel. Lourdes es más baja que Raquel. <i>¿Quién es la más baja?</i> A Elena · B Raquel · C Lourdes	Elena is taller than Julia. Sofia is shorter than Julia. <i>Who is the shortest?</i> A Elena · B Julia · C Sofia	C	Lourdes	C	Sofia
12	Jorge está más feliz que Mario. Jorge está más triste que Raúl. <i>¿Quién es el que está más triste?</i> A Jorge · B Mario · C Raúl	Jorge is happier than Mario. Jorge is less happy than Theo. <i>Who is the saddest?</i> A Jorge · B Mario · C Theo	B	Mario	B	Mario
13	Ángel es más fuerte que Pablo. Pablo es más fuerte que Álvaro. <i>¿Quién es el más fuerte?</i> A Ángel · B Pablo · C Álvaro	Lucas is stronger than Noah. Noah is stronger than Oliver. <i>Who is the strongest?</i> A Lucas · B Noah · C Oliver	A	Ángel	A	Lucas
14	Pablo es más viejo que Miguel. Pedro es más viejo que Pablo. <i>¿Quién es el más joven?</i> A Pablo · B Miguel · C Pedro	Pablo is older than Mike. Pedro is older than Pablo. <i>Who is the youngest?</i> A Pablo · B Mike · C Pedro	B	Miguel	B	Mike

Item	Español	English	ES	Answer	EN	Answer
15	Pilar está más triste que Lucía. Laura está más feliz que Lucía. <i>¿Quién es la más feliz?</i> A Pilar · B Lucía · C Laura	Amelia is meaner than Lana. Laura is nicer than Lana. <i>Who is the nicest?</i> A Amelia · B Lana · C Laura	C	Laura	C	Laura
16	Jorge es más lento que José. David es más rápido que José. <i>¿Quién es el más rápido?</i> A Jorge · B José · C David	Daniel is slower than Samuel. David is faster than Samuel. <i>Who is the fastest?</i> A Daniel · B Samuel · C David	C	David	C	David
17	Isabel es más fuerte que Paula. Sara es más débil que Paula. <i>¿Quién es la más débil?</i> A Isabel · B Paula · C Sara	Ava is stronger than Lisa. Sara is weaker than Lisa. <i>Who is the weakest?</i> A Ava · B Lisa · C Sara	C	Sara	C	Sara
18	David es más rápido que Juan. Juan es más rápido que José. <i>¿David es el más rápido?</i> A David · B Juan · C José	David is faster than Leo. Leo is faster than Jack. <i>Who is the fastest?</i> A David · B Leo · C Jack	A	David	A	David
19	Rocío está más delgada que Julia. Rocío está más delgada que Elena. <i>¿Quién es la que está más delgada?</i> A Rocío · B Julia · C Elena	Lisa is smarter than Julia. Julia is smarter than Elena. <i>Who is the smartest?</i> A Lisa · B Julia · C Elena	A	Rocío	A	Lisa
20	Marta es más débil que Silvia. Ana es más fuerte que Silvia. <i>¿Quién es la más fuerte?</i> A Marta · B Silvia · C Ana	Martha is weaker than Jessica. Anna is stronger than Jessica. <i>Who is the strongest?</i> A Martha · B Jessica · C Anna	C	Ana	C	Anna
21	Mario es más viejo que Aníbal. Miguel es más viejo que Mario. <i>¿Quién es el más viejo?</i> A Mario · B Aníbal · C Miguel	Mario is older than Joe. Milo is older than Mario. <i>Who is the oldest?</i> A Mario · B Joe · C Milo	C	Miguel	C	Milo
22	Mario es más inteligente que Ángel. Mario es más torpe que Jesús. <i>¿Quién es el más torpe?</i> A Mario · B Ángel · C Jesús	Ben is smarter than Liam. Ben is clumsier than Samuel. <i>Who is the clumsiest?</i> A Ben · B Liam · C Samuel	B	Ángel	B	Liam
23	Eva tiene más frío que Belén. Belén tiene más frío que Laura. <i>¿Quién es la que tiene más frío?</i> A Eva · B Belén · C Laura	Eva is sicker than Anna. Anna is sicker than Laura. <i>Who is the sickest?</i> A Eva · B Anna · C Laura	A	Eva	A	Eva
24	Rocío es más joven que Silvia. Rocío es más vieja que Elena. <i>¿Quién es la más joven?</i> A Rocío · B Silvia · C Elena	Evelyn is younger than Emma. Evelyn is older than Sofia. <i>Who is the youngest?</i> A Evelyn · B Emma · C Sofia	C	Elena	C	Sofia
25	Luis es más joven que Aníbal. Aníbal es más joven que Álvaro. <i>¿Quién es el más viejo?</i> A Luis · B Aníbal · C Álvaro	Robert is younger than Charles. Charles is younger than William. <i>Who is the oldest?</i> A Robert · B Charles · C William	C	Álvaro	C	William

Item	Español	English	ES	Answer	EN	Answer
26	Julia es peor que Paula. Lourdes es mejor que Paula. <i>¿Quién es la mejor?</i> A Julia · B Paula · C Lourdes	Maria is meaner than Leah. Sofia is nicer than Leah. <i>Who is the nicest?</i> A Maria · B Leah · C Sofia	C	Lourdes	C	Sofia
27	Pablo está más triste que Ángel. Diego está más triste que Pablo. <i>¿Quién está más triste?</i> A Pablo · B Ángel · C Diego	Pablo is more tired than Alex. Diego is more tired than Pablo. <i>Who is more tired?</i> A Pablo · B Alex · C Diego	C	Diego	C	Diego
28	Eva es más inteligente que Sara. Isabel es más inteligente que Eva. <i>¿Quién es la más torpe?</i> A Eva · B Sara · C Isabel	Eva is smarter than Sara. Isabella is smarter than Eva. <i>Who is the clumsiest?</i> A Eva · B Sara · C Isabella	B	Sara	B	Sara
29	Lucía es más alta que Marta. Lucía es más baja que Ana. <i>¿Quién es la más alta?</i> A Lucía · B Marta · C Ana	Lena is taller than Martha. Lena is shorter than Anna. <i>Who is the tallest?</i> A Lena · B Martha · C Anna	C	Ana	C	Anna
30	Sergio es más joven que Jorge. Sergio es más viejo que José. <i>¿Quién es el más viejo?</i> A Sergio · B Jorge · C José	Brody is younger than James. Brody is older than Noah. <i>Who is the oldest?</i> A Brody · B James · C Noah	B	Jorge	B	James
31	Álvaro es más ligero que Jorge. José es más ligero que Álvaro. <i>¿Quién es el más pesado?</i> A Álvaro · B Jorge · C José	John is faster than George. Robert is faster than John. <i>Who is the slowest?</i> A John · B George · C Robert	B	Jorge	B	George
32	Belén es más fuerte que Pilar. Raquel es más débil que Pilar. <i>¿Quién es la más fuerte?</i> A Belén · B Pilar · C Raquel	Maria is stronger than Susan. Mia is weaker than Susan. <i>Who is the strongest?</i> A Maria · B Susan · C Mia	A	Belén	A	Maria
33	Lidia es más ligera que Pilar. Lidia es más pesada que Isabel. <i>¿Quién es la más ligera?</i> A Pilar · B Lidia · C Isabel	Jennifer is less patient than Maria. Jennifer is more patient than Isabella. <i>¿Who is the least patient?</i> A Jennifer · B Maria · C Isabella	C	Isabel	C	Isabella
34	Lourdes es más joven que Julia. Rocío es más vieja que Julia. <i>¿Quién es la más vieja?</i> A Lourdes · B Julia · C Rocío	Lily is younger than Amelia. Emma is older than Amelia. <i>Who is the oldest?</i> A Lily · B Amelia · C Emma	C	Rocío	C	Emma
35	Miguel está más delgado que Juan. Juan está más delgado que Diego. <i>¿Quién es el que está más delgado?</i> A Miguel · B Juan · C Diego	Nico is more tired than Juan. Juan is more tired than Daniel. <i>Who is the most exhausted?</i> A Nico · B Juan · C Daniel	A	Miguel	A	Nico
36	Ana es más fuerte que Silvia. Ana es más débil que Belén. <i>¿Quién es la más débil?</i> A Ana · B Silvia · C Belén	Anna is stronger than Lisa. Anna is weaker than Mary. <i>Who is the weakest?</i> A Anna · B Lisa · C Mary	B	Silvia	B	Lisa

Item	Español	English	ES	Answer	EN	Answer
37	Carlos es más inteligente que Pablo. Carlos es más torpe que Álvaro. <i>¿Quién es el más inteligente?</i> A Carlos · B Pablo · C Álvaro	Carlos is smarter than Pablo. Alvaro is smarter than Carlos. <i>Who is the smartest?</i> A Carlos · B Pablo · C Alvaro	C	Álvaro	C	Alvaro
38	Ángel es más alto que Miguel. Miguel es más alto que Diego. <i>¿Quién es más bajo?</i> A Ángel · B Miguel · C Diego	Mateo is taller than Miguel. Miguel is taller than Diego. <i>Who is the shortest?</i> A Mateo · B Miguel · C Diego	C	Diego	C	Diego
39	José es mejor que Jesús. Aníbal es peor que Jesús. <i>¿Quién es el mejor?</i> A José · B Jesús · C Aníbal	Henry is more generous than Juan. John is less generous than Juan. <i>Who is the most generous?</i> A Henry · B Juan · C John	A	José	A	Henry
40	Paula tiene más frío que Raquel. Raquel tiene más frío que Laura. <i>¿Quién es la que tiene más frío?</i> A Paula · B Raquel · C Laura	Maria is colder than Emma. Emma is colder than Laura. <i>Who is the coldest?</i> A Maria · B Emma · C Laura	A	Paula	A	Maria
41	David es más ligero que Juan. Luis es más ligero que David. <i>¿Quién es el más pesado?</i> A David · B Juan · C Luis	David is more patient than Juan. Luis is more patient than David. <i>Who is the least patient?</i> A David · B Juan · C Luis	B	Juan	B	Juan
42	César es más afectuoso que José. Jorge es más afectuoso que César. <i>¿Quién es más afectuoso?</i> A César · B José · C Jorge	Lucas is more caring than Daniel. George is more affectionate than Lucas. <i>Who is the most affectionate?</i> A Lucas · B Daniel · C George	C	Jorge	C	George
43	Lucía es más débil que Marta. Lucía es más fuerte que Eva. <i>¿Quién es la más fuerte?</i> A Lucía · B Marta · C Eva	Lola is weaker than Martha. Lola is stronger than Eva. <i>Who is the strongest?</i> A Lola · B Martha · C Eva	B	Marta	B	Martha
44	Ángel es más ligero que Aníbal. Luis es más pesado que Aníbal. <i>¿Quién es más pesado?</i> A Ángel · B Aníbal · C Luis	David is quieter than Robert. Luis is louder than Robert. <i>Who is the loudest?</i> A David · B Robert · C Luis	C	Luis	C	Luis
45	Juan es más joven que Miguel. Jesús es más joven que Juan. <i>¿Quién es el más joven?</i> A Juan · B Miguel · C Jesús	Juan is younger than Miguel. John is younger than Juan. <i>Who is the youngest?</i> A Juan · B Miguel · C John	C	Jesús	C	John
46	Elena es más rápida que Paula. Lucía es más lenta que Paula. <i>¿Quién es la más lenta?</i> A Elena · B Paula · C Lucía	Elena is faster than Pedro. Luisa is slower than Pedro. <i>Who is the slowest?</i> A Elena · B Pedro · C Luisa	C	Lucía	C	Luisa
47	Isabel es más joven que Pilar. Pilar es más joven que Laura. <i>¿Quién es la más vieja?</i> A Isabel · B Pilar · C Laura	Diana is taller than Robert. David is taller than Diana. <i>Who is the tallest?</i> A Diana · B Robert · C David	C	Laura	C	David

Item	Español	English	ES	Answer	EN	Answer
48	Mario es más alto que Álvaro. David es más alto que Mario. <i>¿Quién es el más bajo?</i> A Mario · B Álvaro · C David	James is taller than Charles. Andy is taller than James. <i>Who is the shortest?</i> A James · B Charles · C Andy	B	Álvaro	B	Charles
49	Ana es más alta que Noelia. Noelia es más alta que Marta. <i>¿Quién es la más alta?</i> A Ana · B Noelia · C Marta	Anna is louder than Olivia. Olivia is louder than Maya. <i>Who is the loudest?</i> A Anna · B Olivia · C Maya	A	Ana	A	Anna
50	Belén es más lenta que Raquel. Raquel es más lenta que Rocío. <i>¿Quién es la más rápida?</i> A Belén · B Raquel · C Rocío	Sophia is slower than Mia. Mia is slower than Ava. <i>Who is the fastest?</i> A Sophia · B Mia · C Ava	C	Rocío	C	Ava

SECTION 07

Numerical Speed

Velocidad Numérica · 3 practice + 60 scored · 4 minutes

Instruction · Spanish	¿Cuál número está más lejos del valor medio?
Instruction · English	Which number is farther from the middle number?

The key is the **position** of the number farthest from the median — **A** the first, **B** the second, **C** the third — with **that number** beside it.

Item	A B C	Key	Is	Item	A B C	Key	Is	Item	A B C	Key	Is
P1	7 1 3	A	7	19	10 25 18	A	10	40	13 25 20	A	13
P2	21 29 17	B	29	20	26 18 12	A	26	41	21 14 9	A	21
P3	5 9 25	C	25	21	5 10 18	C	18	42	27 14 20	A	27
1	6 11 17	C	17	22	17 9 2	A	17	43	21 14 6	C	6
2	15 1 7	A	15	23	3 14 8	B	14	44	18 24 11	C	11
3	12 19 7	B	19	24	10 21 16	A	10	45	26 11 18	A	26
4	20 5 13	B	5	25	21 16 9	C	9	46	9 1 14	B	1
5	13 18 26	C	26	26	23 15 10	A	23	47	16 21 10	C	10
6	2 15 10	A	2	27	2 8 15	C	15	48	16 21 27	C	27
7	15 9 20	B	9	28	28 13 21	B	13	49	13 21 27	A	13
8	12 19 24	A	12	29	13 18 5	C	5	50	9 1 14	B	1
9	5 20 12	B	20	30	23 15 29	B	15	51	14 20 27	C	27
10	4 11 17	A	4	31	15 30 22	B	30	52	16 9 4	A	16
11	7 14 1	B	14	32	20 6 12	A	20	53	3 9 17	C	17
12	28 16 23	B	16	33	18 25 13	B	25	54	23 18 12	C	12
13	27 20 15	A	27	34	15 10 21	C	21	55	19 13 8	A	19
14	15 27 22	A	15	35	12 23 18	A	12	56	25 10 18	B	10
15	22 14 27	B	14	36	16 29 23	A	16	57	21 13 8	A	21
16	29 16 22	A	29	37	11 24 16	B	24	58	24 29 17	C	17
17	23 29 16	C	16	38	20 6 12	A	20	59	2 17 9	B	17
18	30 24 16	C	16	39	18 4 12	B	4	60	4 17 11	A	4

SECTION 08

Working Memory

Memoria de Trabajo · 3 practice + 60 scored · 4 minutes

Instruction · Spanish	¿Cuál letra exterior está más lejos alfabéticamente de la letra del centro?
Instruction · English	Which outer letter is alphabetically farther from the middle letter?

The key is **left** or **right**, according to which outer letter lies farther in the alphabet from the middle one, with **that letter** beside it.

Item	Letters	Key	Is	Item	Letters	Key	Is	Item	Letters	Key	Is
P1	F H K	right	K	19	G K M	left	G	40	J K R	right	R
P2	P S U	left	P	20	K M Q	right	Q	41	R U W	left	R
P3	C E H	right	H	21	R U Y	right	Y	42	Q S V	right	V
1	K N R	right	R	22	R V Y	left	R	43	M P T	right	T
2	G K N	left	G	23	U X Z	left	U	44	L N R	right	R
3	Q V Y	left	Q	24	C H K	left	C	45	G H J	right	J
4	D G L	right	L	25	G J O	right	O	46	O Q S	left	O
5	J L Q	right	Q	26	N P U	right	U	47	K M T	right	T
6	J Q R	left	J	27	K N S	right	S	48	F H L	right	L
7	D F I	right	I	28	Q T X	right	X	49	J N P	left	J
8	J N P	left	J	29	Q U X	left	Q	50	Q R T	right	T
9	B D H	right	H	30	E J N	left	E	51	L M Q	right	Q
10	R U W	left	R	31	D H J	left	D	52	O P U	right	U
11	M P U	right	U	32	Q T V	left	Q	53	T V X	left	T
12	M R V	left	M	33	N S U	left	N	54	G J L	left	G
13	F K M	left	F	34	G L O	left	G	55	R S U	right	U
14	N S W	left	N	35	I L P	right	P	56	P Q R	left	P
15	E G L	right	L	36	N O R	right	R	57	M N T	right	T
16	S U X	right	X	37	F G K	right	K	58	K L P	right	P
17	Q V X	left	Q	38	P Q S	right	S	59	Q S V	right	V
18	O S X	right	X	39	T W X	left	T	60	O R U	left	O

SECTION 09

Visual Rotation

Rotación Visual · 3 practice + 60 scored · 5 minutes

Instruction · Spanish	¿Cuántas columnas tienen figuras iguales? (rotaciones permitidas, espejos no)
Instruction · English	How many columns contain the same figure? (rotations allowed, mirrors not)

The figures are printed **exactly as the student sees them**. Under each item they are repeated in notation: R is a normal R and M a mirrored R, each followed by its rotation in degrees. The **Matches** line says which columns match, which is where the key comes from. Remember that **only handedness counts**.

Practice items (not scored)

P1	Key 1	P2	Key 1	P3	Key 1
					

Scored items

1	Key 1	2	Key 2	3	Key 3	4	Key 1
							
5	Key 1	6	Key 0	7	Key 1	8	Key 2
							

9 Key 1 Б Р Б Р Р Р M270 R270 M180 R270 R90 R90 Matches: col. 2	10 Key 2 Б Я Я Б Б Я
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M270 M90 R180
M180 M180 M90
Matches: col. 1, 2

29 Key 1       M270 R180 M90 R180 M270 M270 Matches: col. 3	30 Key 1       R180 R270 R90 M90 R180 M180 Matches: col. 2	31 Key 3       M90 M180 M270 M270 M270 M180 Matches: col. 1, 2, 3	32 Key 3       M90 R180 R90 M270 R270 R270 Matches: col. 1, 2, 3
33 Key 3       R180 R90 M90 R270 R270 M180 Matches: col. 1, 2, 3	34 Key 3       M180 M270 M90 M90 M90 M180 Matches: col. 1, 2, 3	35 Key 0       R90 R180 M90 M90 M90 R90 Matches: none	36 Key 1       R90 M180 R270 R270 R90 M180 Matches: col. 1
37 Key 2       R180 M90 R90 R270 M180 M90 Matches: col. 1, 2	38 Key 2       M270 M180 M90 M180 R90 M180 Matches: col. 1, 3	39 Key 0       R180 M90 R90 M90 R180 M180 Matches: none	40 Key 3       M180 M270 M90 M90 M180 M180 Matches: col. 1, 2, 3
41 Key 0       R90 R180 M90 M90 M90 R180 Matches: none	42 Key 2       M90 M180 R90 R180 M270 R180 Matches: col. 2, 3	43 Key 0       R90 R180 M90 M90 M270 R90 Matches: none	44 Key 2       R90 R180 M90 M180 R270 M270 Matches: col. 2, 3
45 Key 0       M90 M180 R90 R90 R90 M180 Matches: none	46 Key 2       M90 M180 M270 R90 M90 M90 Matches: col. 2, 3	47 Key 3       M180 M270 M90 M270 M180 M270 Matches: col. 1, 2, 3	48 Key 1       M180 R270 R90 R90 R180 M90 Matches: col. 2

49 Key 1 К К Я Я К Я M180 M270 M90 R90 M180 R90 Matches: col. 2	50 Key 1 Я Я Я Я Я К
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R180 R90 M90
M90 M90 M180

SECTION 10

Answer key distribution

How balanced the bank is

Pattern Recognition — 60 scored items

0	1	2	3	4
10 · 16%	14 · 23%	12 · 20%	12 · 20%	12 · 20%

Verbal Reasoning — 50 scored items

A	B	C
10 · 20%	12 · 24%	28 · 56%

Numerical Speed — 60 scored items

A	B	C
25 · 41%	18 · 30%	17 · 28%

Working Memory — 60 scored items

left	right
27 · 45%	33 · 55%

Visual Rotation — 60 scored items

0	1	2	3
14 · 23%	16 · 26%	13 · 21%	17 · 28%

Working Memory leans slightly towards **right** (35 of 63) and Numerical Speed towards **A** (26 of 63). Neither is extreme, but it is worth knowing if the bank is ever extended or resampled: a student who always gave the same answer would not land at chance.

SECTION 11

Cautions

What to know before administering it

Resolved — the instructions and warnings are bilingual now

Until 24 September 2026, four subtests gave their instruction in Spanish only, and the 30- and 10-second warnings appeared in Spanish for everyone. No longer. Outstanding: a **native-speaker review** of the English wordings.

Portuguese silently resolves to English

There is no Portuguese version, and a browser set to Portuguese resolves to English with no warning to the student or to whoever is administering the test. If it is added, do it **per subtest**: three need only the instruction line, **Working Memory needs an item review** (19 of 63 use K, W or Y) and Verbal Reasoning needs 53 new items. See section 04.

Editing an item moves its answer

Four of the five subtests compute the key from the item's data. Regenerate this document after any change to the bank.

Ñ moves one Working Memory item, today, in Spanish

The scorer orders the alphabet A=1 ... Z=26, with no Ñ. A student reciting the Spanish alphabet (27 letters) measures differently from O onwards. On **1 of the 63 scored items** that changes the perceived answer. It is small, but it is real and it is already in the field.

What decides a match is handedness, not rotation

Worth repeating to anyone checking Visual Rotation by eye: two normal Rs at different angles DO match. It is the most common mistake made when reviewing.